

Quantifying Market Power and Business Dynamism in the Macroeconomy

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Motivation and Research Question

Active debate about the rise in mark-ups, fall in business dynamism and fall in labor share, productivity slowdown...

The literature has put forth several factors at the origin for these facts:

- ▶ Changes to barriers to entry
- ▶ Changes to overhead costs or span-of-control costs
- ▶ Decrease in innovation, Idea Diffusion ...

This paper asks the following questions

1. What are underlying forces at the origin of the secular transformation of the macro-economy between 1980-2016
2. How to identify and measure the respective contribution of each separate force?
3. How to disentangle the welfare consequences of each of these changes?

What this paper does

Through the lens of a model, estimate which structural forces are driving recent macroeconomic trends.

Model (Structural origins)

1. Market Structure

Number of potential entrants M

2. Technology

Dispersion in firm-level productivity σ

3. Entry Costs

Overhead costs associated with being active Φ



Data (Secular Trends)

1. Mark-ups

Rise in average sales-weighted μ

2. Share of fixed costs

Rise in the share of overhead in total costs

3. Labor reallocation (business dynamism)

Decrease in reallocation from entry, exit and incumbents

Idea: Secular Trends are endogenous objects and may mask potentially opposing forces at play. For instance,

- ▶ Higher entry costs $\rightarrow$ less players, more market power
- ▶ Higher entry costs $\rightarrow$ Only the most productive firms are remaining, less players have to "waste" overheads

Preview of the Results

- ▶ The 3 structural forces are necessary to replicate recent observed trends in the macroeconomy
- ▶ Mainly, the decrease in number of potential entrants and the rise in entry costs seem to explain most of the changes
- ▶ There are positive and negative effects from these secular transformations...
 - ▶ It led to an increase in aggregate mark-ups and share of overhead costs, estimated to lower Welfare by 15 percent...
 - ▶ ... While also led to increased productivity and increased selection of most efficient firms, estimated to increase welfare by 6 percent

Model Overview

Estimation Strategy

Welfare Analysis

Model environment

Static model of oligopolistic competition in the product market

Atkeson and Burstein (2008)

- ▶ Two-level nested CES with strategic interactions within sectors
 - ▶ **Sectors:** Continuum $j \in [0, 1]$ with elasticity of substitution θ
 - ▶ **Firms:** discrete nb i in $\{1, \dots, M_{jt}\}$ in each small market j , with elasticity of substitution θ

With an entry-step akin to **Berry (1992)**

- ▶ Up to M potential entrants
- ▶ Firms prefer in some periods to stay inactive due to a fixed overhead cost of producing, $M_{jt} \leq M$

Household problem

- ▶ Infinitely-lived representative consumer with GHH preferences

$$U_0 = \sum_{t=0}^{\infty} \beta^t U \left(C_t - \bar{\varphi}^{-\frac{1}{\varphi}} \frac{N_t^{1+\frac{1}{\varphi}}}{1 + \frac{1}{\varphi}} \right)$$

- ▶ Aggregate consumption C_t is some CES aggregate of individual firm varieties. Love-for-variety is neutralized.

$$C_t = \left(\int_0^1 c_{jt}^{\frac{\theta-1}{\theta}} dj \right)^{\frac{\theta}{\theta-1}} \quad c_{jt} = \left[\sum_{i=1}^{M_{jt}} M_{jt}^{-\frac{1}{\eta}} c_{ijt}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}$$

Solution

- ▶ Gives rise to standard labor supply condition (no wealth effect)

$$N_t = \bar{\varphi} W_t^{\varphi}$$

- ▶ and input demand condition

$$y_{ijt} = \left(\frac{p_{ijt}}{p_{jt}} \right)^{-\eta} \left(\frac{p_{jt}}{P_t} \right)^{-\theta} C_t$$

Individual Firm problem

- ▶ Linear technology with labor as a single input

$$y_{ijt} = z_{ijt} n_{ijt}$$

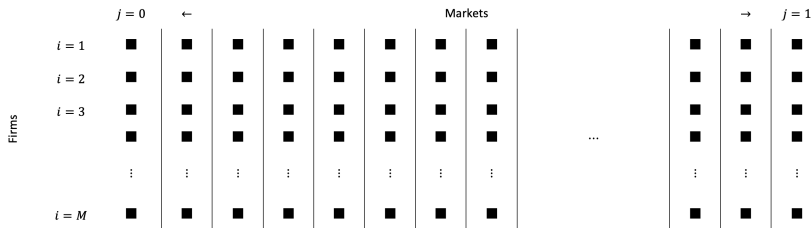
- ▶ Technology z_{ijt} is **exogenous** and evolves according to an AR(1) process in logs:

$$\log z_{ijt} = \rho \log z_{ijt-1} + \varepsilon_{ijt} \quad , \quad \varepsilon_{ijt} \sim \mathcal{N}\left(-\frac{1}{2} \frac{\sigma^2}{1 + \rho}, \sigma^2\right)$$

- ▶ If the firm decides to be active, it must pay a fixed overhead cost Φ in terms of labor

$$\pi_{ijt} = p_{ijt} y_{ijt} - W_t n_{ijt} - W_t \Phi$$

Industry Equilibrium



- Strategic competition within sector j , as in Atkeson and Burstein (2008)
- Set prices internalizing the effect on p_{ijt} and y_{jt} . Take aggregate as given.

$$y_{ijt} = \left(\frac{p_{ijt}}{p_{jt}} \right)^{-\eta} \left(\frac{p_{jt}}{P_t} \right)^{-\theta} C_t$$

- Suppose there are $M_{jt} \leq M$ active in sector j , then solve a static Nash Equilibrium within sector j to get pricing decisions.

Industry Equilibrium

- In the case of quantity competition (Bertrand), optimal price is:

$$p_{ijt} = \underbrace{\frac{\omega_{ijt}}{\omega_{ijt} - 1}}_{=\mu_{ijt}} W_t, \quad \omega_{ijt} = [s_{ijt}\theta^{-1} + (1 - s_{ijt})\eta^{-1}]^{-1}$$

- Solved numerically as a fixed-point problem
 1. Given a distribution of technologies $z_{jt} = (z_{1jt}, \dots, z_{M_{jt}})$, take a guess for the distribution of mark-ups $\mu_{jt} = (\mu_{1jt}, \dots, \mu_{M_{jt}})$
 2. Compute demand elasticity ε_{jt} , market shares s_{jt} as closed form of the guess and technology.
 3. Update on μ_{jt} until convergence
- How to solve for the cut-off technology $\overline{z}_{jt}$ such that firms with higher productivity want to enter?

$$\left(\frac{\mu_{ijt}}{W_t} z_{ijt} - W_t \right) n_{ijt} \geq W_t \Phi,$$

with equality for $\overline{z}_{jt}$

Aggregation and Market Clearing

- ▶ By LLN, distribution of vector z_j across j equal to the the density of the technology process (hence is stationary)
- ▶ Cutoffs $\bar{z}_j$'s will depend on aggregates $Y, N, W...$ (another loop)
- ▶ **Labor Market Clearing:** N and W are determined jointly

$$\underbrace{\bar{\varphi} W_t^\varphi}_{=N^s(W)} = \underbrace{\int_0^1 \sum_{i=1}^M \mathbb{I}\{z_{ij} \geq \bar{z}_j\} (n_{ij} + \phi) dj}_{=N^d(W)=N^{prod}+\Phi}$$

- ▶ For the aggregate output Y , we can solve for the aggregate mark-up μ that gives **Goods Market clearing**.

$$PY = \int_0^1 \sum_{i=1}^M \mathbb{I}\{z_{ij} \geq \bar{z}_j\} p_{ij} y_{ij} dj \quad \Rightarrow \quad \mu = \frac{Z}{W/P}$$

Entry condition

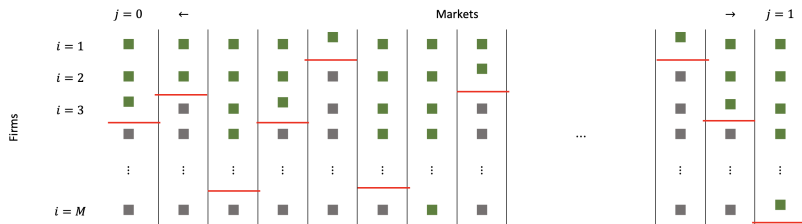
Starting with an equilibrium where all firms enter, suppose some are not profitable ($\pi_{ijt} < 0$ due to fixed costs)...

	Markets											
	$j = 0$	←								→	$j = 1$	
$i = 1$	■	■	■	■	■	■	■	■	■	■	■	
$i = 2$	■	■	■	■	■	■	■	■	■	■	■	
$i = 3$	■	■	■	■	■	■	■	■	■	■	■	
	■	■	■	■	■	■	■	■	■	■	■	
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	
$i = M$	■	■	■	■	■	■	■	■	■	■	■	

...

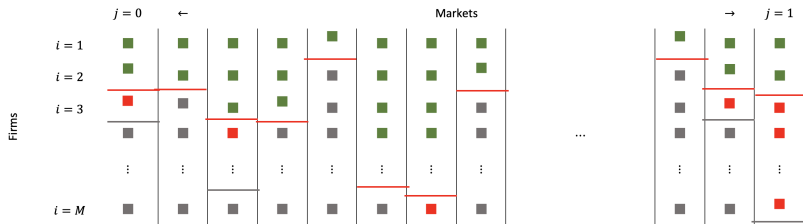
Entry condition

Resolve the industry equilibrium and aggregate quantities Y, N, W shutting off these firms...



Entry condition

Do this until the set of incumbents is stable...



Procedure similar as Berry (1992).

Analysing Distortions from Market Power

NB: In this paper, Z is NOT the TFP... $Y \neq ZN^{prod}$

$$Z := \left[\int_0^1 (z_j)^{\theta-1} dj \right]^{\frac{1}{\theta-1}}, \quad z_j = \left[\frac{1}{\sum_{i=1}^M \mathbb{I}\{z_{ij} \geq \bar{z}_j\}} \sum_{i=1}^M \mathbb{I}\{z_{ij} \geq \bar{z}_j\} z_{ij}^{\eta-1} \right]^{\frac{1}{\eta-1}}$$

- ▶ Expression akin to the expression for TFP in the Planner's Economy Z^{SP} . However, cutoffs for entry will be different $\bar{z}_j^{SP} \neq \bar{z}_j$
- ▶ Hence, μ is NOT a cost-weighted arithmetic average or a sales-weighted harmonic average of firm-level markups as in Edmond et al. (2015)

$$\mu = \frac{Z}{W/P} = \left[\int_0^1 \left(\frac{z_j}{Z} \right)^{\theta-1} \mu_j^{1-\theta} \right]^{\frac{1}{1-\theta}}$$
$$\mu_j = \left[M_j^{-1} \sum_{i=1}^{M_j} \left(\frac{z_{ij}}{z_j} \right)^{\eta-1} \mu_{ij}^{1-\eta} \right]$$

Analysing Distortions from Market Power

Whole idea is to rewrite deviation of output Y from a given benchmark

- ▶ Benchmark is NOT the social-planner allocation!
- ▶ Benchmark is the level that satisfies $\tilde{Y} = ZN^{prod}$ (the "undistorted" output)

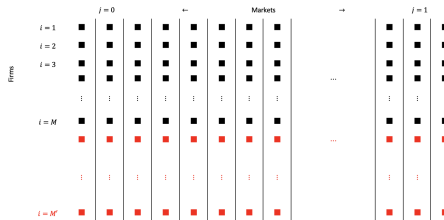
$$N^d = \underbrace{\frac{Y}{Z}}_{\text{Benchmark}} \underbrace{\left(\mu \frac{W/P}{Z} \right)^{-\theta}}_{\text{Agg. Mkup dist.}} \underbrace{\Omega}_{\text{Mkup disp.}} + \underbrace{\Phi}_{\text{Fixed Costs}}$$

- ▶ Later, benchmark is the deviation of output from another abstract benchmark $Y = Z^* N^{tot}$ due to 5 wedges:
 1. Irrespective of entry decisions, agg. productivity may vary (Z_t^*)
 2. Due to the entry decision, there may be selection of more or less productive firms ($S_t = \frac{Z_t}{Z_t^*}$)
 3. More or less labor can be "wasted in overhead" $\tilde{\Phi} = \left(\frac{N}{N-\Phi} \right)^\alpha$
 4. There is an aggregate mark-up μ_t creating a wedge between MPL and wage
 5. There is misallocation due to mark-up dispersion

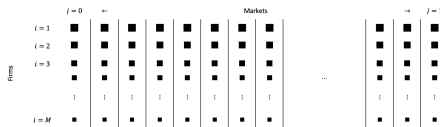
No explicit algebra, but definitely not wedges of decentralized vs. social allocation!

Structural Parameters Estimated

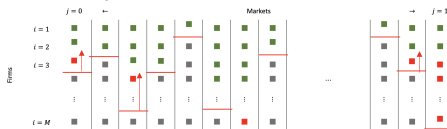
1. Market structure M : nb of potential entrants



2. Technology σ : more dispersed productivity



3. Fixed cost (overhead) ϕ : more expensive to be active



Also Between- θ and Within-sector η CES

Data Moments Selected

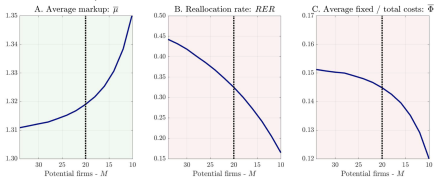


- A. Sales weighted average markup (Eeckhout, De Loecker, Unger, 2020)
- B. Annual employment reallocation rate (Census, BDS)
- C. Total Sales, Goods and Administrative expenses to Total costs (Compustat)

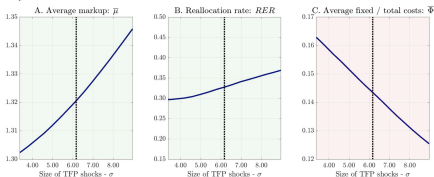
- ▶ Model augmented for capital and other intermediary inputs (be closer to Compustat) ▶ Moment Definition
- ▶ All moments are sales-weighted
- ▶ Estimate $\{M_t, \sigma_t, \phi_t\}_{t=1980}^{2016}$ for every single year $\rightarrow$ get a time series of each parameter
- ▶ CES parameters θ and η only used for pre-1995 targets (under-identification pre-1995...)
- ▶ Rest of parameters is calibrated ▶ Calibration

Moments not observationally equivalent from Model

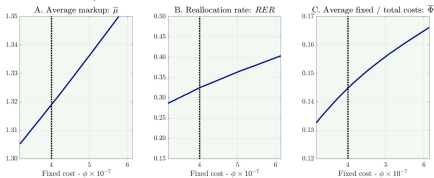
1. Market structure $M \downarrow$:



2. Technology $\sigma \uparrow$:



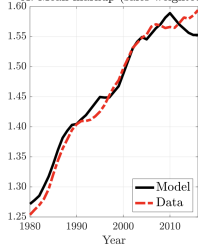
3. Fixed cost (overhead) $\phi \uparrow$:



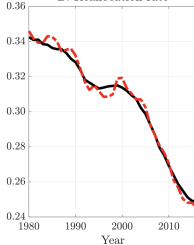
Estimation Results

Moment Matchings

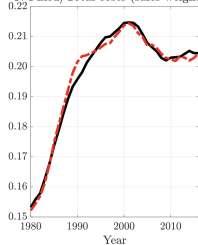
A. Mean markup (sales weighted)



B. Reallocation rate

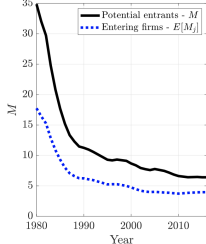


C. Fixed/Total costs (sales weighted)

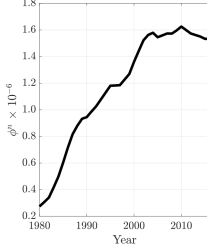


Parameters Estimates

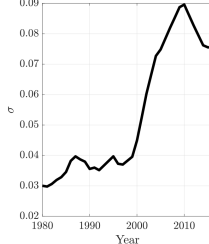
A. Potential entrants - M_t



B. Fixed cost - ϕ_t

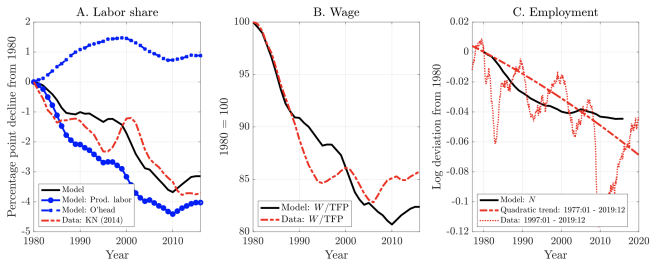


C. Productivity shock size - σ_t



Validation Procedure

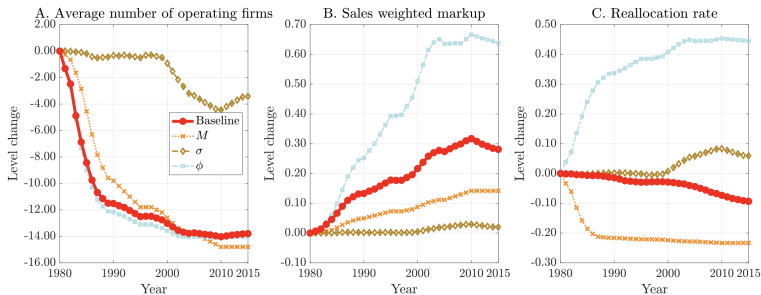
Match other moments not targeted in the estimation...



- ▶ Also look at job entry and exit separately ▶ Entry/exit
- ▶ As well as job reallocation between small and large firms
▶ Firm size
- ▶ And mark-up dispersion ▶ Mkup disp

Use the Model as a Laboratory....

- See what would have happened if only one parameter had varied



- No model parameter alone can explain the secular changes in the macroeconomy
- At the forefront, changes in potential number of firms M and Fixed costs ϕ drive most of the secular trend

Wedges and Positive Analysis

Idea behind the paper is that there could be "good" and "bad" outcomes from changes in the structural parameters...

Parameters $\Rightarrow$ **Wedges** $\Rightarrow$ **Output & Welfare**

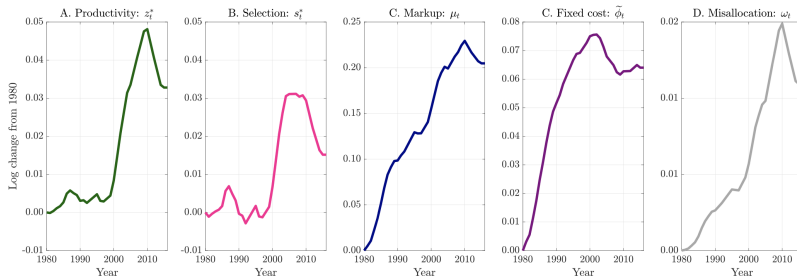
Wedges

1. **Latent productivity** Z_t^* : Irrespective of entry decisions, aggregate productivity may vary
2. **Selection** $S_t = \frac{Z_t}{Z_t^*}$: due to entry decisions, selection of more or less productive firms
3. **Fixed costs** $\tilde{\Phi} = (\frac{N}{N-\Phi})^\alpha$: More or less labor can be "wasted in overhead"
4. **Aggregate mark-up** μ_t : creates a wedge between MPL and wage
5. **Misallocation** Ω : due to mark-ups dispersion

Undervived log-deviation expansion shows that:

$$y_t = \left(\frac{1+\varphi}{\alpha} \right) \left(\underbrace{Z_t^*}_{\text{Latent prod.}} + \underbrace{S_t}_{\text{Select.}} \right) - \left(\frac{1+\varphi-\alpha}{\alpha} \right) \underbrace{\mu_t}_{\text{Agg. Mark-up}} - \frac{1}{\alpha} \underbrace{\tilde{\phi}_t}_{\text{Fixed costs}} - \underbrace{\omega_t}_{\text{Misalloc.}}$$

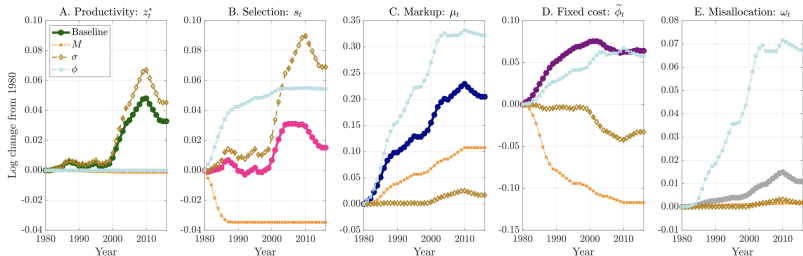
Each wedge witnessed a secular change...



- Primer decomposition of the "good" and "bad" effects of market power

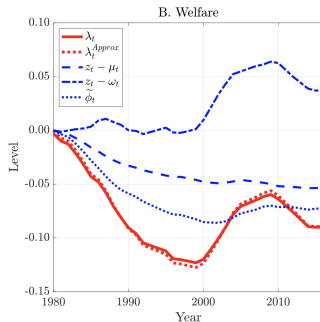
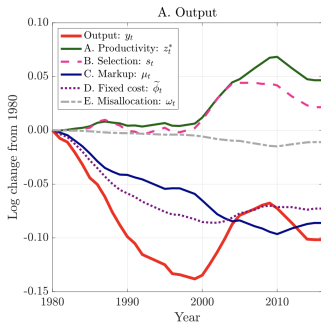
Decomposition of wedges

How is each structural parameter affecting the wedges?



- Structural parameters may inequally affect each of the wedges

Decomposition of Output & Welfare changes



► For Output,

$$y_t = \left(\frac{1 + \varphi}{\alpha} \right) \left(\underbrace{z_t^*}_{\text{Latent prod.}} + \underbrace{s_t}_{\text{Select.}} \right) - \left(\frac{1 + \varphi - \alpha}{\alpha} \right) \underbrace{\mu_t}_{\text{Agg. Mark-up}} - \frac{1}{\alpha} \underbrace{\tilde{\phi}_t}_{\text{Fixed costs}} - \underbrace{\omega_t}_{\text{Misalloc.}}$$

► For Welfare, consider the cons.-equivalent change from baseline

$$\lambda_t = \left(\frac{1 - \alpha + (1 - ls)\varphi}{\alpha} \right) \left(\underbrace{z_t^* + s_t}_{=z_t} - \mu_t \right) + (z_t - \omega_t) - \frac{1}{\alpha} \tilde{\phi}_t$$

Conclusion and Comments

- ▶ From the lens of a model, explain the rise in mark-ups and overheads + drop in labor reallocation based on 3 "deep" forces
- ▶ Quantify the importance of each force and its relative impact on output and welfare
- ▶ Unveil the opposing trends at play, with market power having potentially good and bad effects

My comments

- ▶ Great insights from doing "GMM"
- ▶ Why focus on these 3 structural forces in particular? What do we actually learn? (what is M ?) Do we have existing evidence on each on them from the literature?
- ▶ Positive and Normative analysis needs to be with respect to an efficient benchmark, seems "ad-hoc" currently...
- ▶ Nothing is dynamic: the evolution of technology is fully deterministic so this rules out interesting mechanisms such as entry deterrence motive, collusion, ... Isn't this the real tradeoff between Efficiency and Market Power?

Appendix - Calibration

▶ Back

Parameter		Value	Source
Labor Supply Elasticity	φ	0.25	Chetty et al. (2011)
Discount rate	β	0.96	Real interest rate 4% p.a.
Depreciation rate	δ	0.12	De Loecker et al. (2020)
Productivity persistence	ρ	0.90	Decker et al. (2017)
Factor share: Labor plus intermediates in variable cost	$\alpha^{COGS} = 1 - \alpha^k$	0.88	Compustat data
Factor share: Labor in labor plus intermediates	$\psi = \psi^{COGS} = \psi^{SGA}$	0.33	Compustat data

For CES parameters,

$$\theta = \widehat{1.20} \qquad \eta = \widehat{5.75}$$

Appendix - Moments Definition (Model)

► Back

A. Sales weighted average markup

$$\bar{\mu} := \int s_j \sum_{i=1}^{M_j} s_{ij} \mu_{ij} dj$$

B. Employment reallocation rate

$$RER := \frac{JC + JD}{N} \quad , \quad JC := \int \sum_{i=1}^{M_j} \left(n'_{ij} - n_{ij} \right)^+ dj$$

C. Fixed cost share

$$\bar{\Phi} := \frac{WN^{Ohead}}{WN^{Ohead} + WN^{Prod}}$$

Appendix - Moments Definition (Data)

► Back

1. **Markup.** The measure of markups in terms of the data (as discussed in Appendix B) is given by: $\mu_{it} = \alpha^{COGS} / (COGS_{it} / Sales_{it})$, where optimality implies that the output elasticity α^{COGS} can be measured from the share of costs of goods sold in total variable costs:

$$\alpha^{COGS} = \frac{COGS_{it}}{COGS_{it} + CapitalCosts_{it}}.$$

The expression from the model is equation (B5).

2. **Fixed costs.** For our measure of fixed to total costs we use the data to compute the ratio of total SGA_{it} to $TotalCosts_{it}$, which in the model is:

$$\frac{SGA_{it}}{TotalCosts_{it}} = \frac{\phi}{n_{it}/\alpha^{COGS} + \phi}.$$

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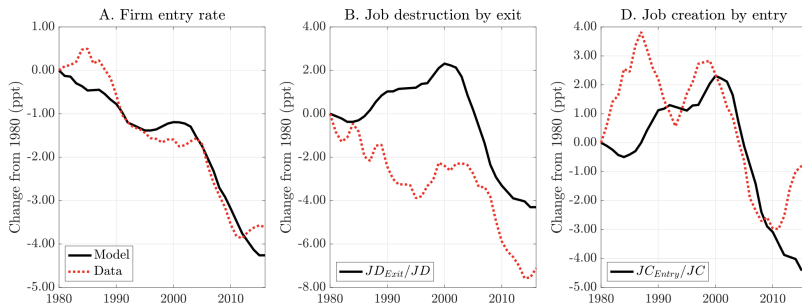
3. **Reallocation rate.** The reallocation rate is constructed using changes in total employment ($n_{it} + \phi$), which gives

$$RER_t = \frac{\sum_i (n_{it+1} - n_{it})^+ + \sum_i (n_{it} - n_{it+1})^-}{\frac{1}{2} (\sum_i n_{it} + \sum_i n_{it+1}) + M_t \phi}.$$

Appendix - More on Job reallocation

► Back

Validation II - Decline in entry and composition of job creation / destruction

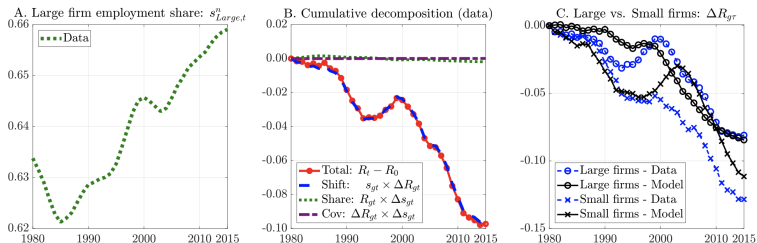


- Two forces: Higher ϕ , Higher σ , but reduction in density around cut-off

Appendix - Job reallocation between small and large firms

► Back

Validation III - Reallocation rates in the cross-section



$$\underbrace{R_t - R_0}_{Total_t} = \underbrace{\sum_{\tau=1}^t \sum_{g=1}^G s_{g\tau} \times [\Delta R_{g\tau}]}_{Shift_t \approx 95\%} + \underbrace{\sum_{\tau=1}^t \sum_{g=1}^G R_{g\tau} \times [\Delta s_{g\tau}^n]}_{Share_t \approx 5\%} + \underbrace{\sum_{\tau=1}^t \sum_{g=1}^G [\Delta s_{g\tau}^n] \times [\Delta R_{g\tau}]}_{Covariance_t \approx 0\%} \quad (1)$$

Appendix - More on Mark-ups

► Back

Validation IV - Sales reallocation and average markup

$$\Delta \bar{\mu}_t = \underbrace{\sum_i \mu_{i,t-1} \Delta m_{i,t} + \sum_i \Delta \mu_{i,t} \Delta m_{i,t}}_{\text{(i) } \Delta \text{ Reallocation of incumbent market shares}} + \underbrace{\sum_i m_{i,t-1} \Delta \mu_{i,t}}_{\text{(ii) } \Delta \text{ Within firm markup changes}} + \underbrace{\sum_{i \in \text{Entry}} \mu_{i,t} m_{i,t} - \sum_{i \in \text{Exit}} \mu_{i,t-1} m_{i,t-1}}_{\text{(iii) Net entry}}$$

